Spontaneous Symmetry Breaking

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1 Symmetries and Noether theorem

A state is said top be symmetric under some transformation if the transformed state cannot be recognized as different from the initial one. In quantum mechanics, states are defined as subspaces of the Hilbert spaces, therefore a state is identified up to a multiplication (complex) constant; the normalization condition $\langle\psi|\,|\psi\rangle = 1$ does not remove completely this freedom, leaving a global phase ambiguity: we can reformulate this statement as a postulate intrinsic in quantum mechanics, which tells that the total phase of a quantum state cannot is not measurable. If $|\psi\rangle$ is a quantum state and U a transformation – that can in principle act on spacetime coordinates or other internal degrees of freedom – we say that U is a symmetry if $U\,|\psi\rangle = e^{i\phi}\,|\psi\rangle$. In the Schrödinger non-relativistic formulation of quantum mechanics, the dynamic equation that specifies completely the evolution of a quantum system depends exclusively on the Hamiltonian of the system, therefore a symmetry in the equation of motions of the system is directly linked to a symmetry in the Hamiltonian operator H. If the expectation value of H is the same between different states $|\psi\rangle$ and $U\,|\psi\rangle$,

i.e. $\langle\psi|H|\psi\rangle = \langle\psi|U^\dagger HU|\psi\rangle$, we can conlcude either that the state is symmetric under the transformation U or that $U^\dagger HU = H$. In the last case we say that the Hamiltonian is symmetric under the transformation U or, equivalently, that $[U, H] = 0$.

From now, we consider only unitary transformations, so that the normalization condition is respected even for the transformed state. A unitary operator can be always be seen as the exponential of an anti-hermitian operator, or equivalently $U = e^{iQ}$, where Q is a hermitian operator. We can easily demonstrate that if U is a symmetry of the Hamiltonian, then commutes also with the time evolution operator e^{iHt} and therefore the expectation value of Q on any state does not depend on the time evolution of the state: $\langle Q\rangle$ is a conserved quantity. The existence of symmetries seems to bring naturally conservation lasws, following this idea, we expect that continuous symmetries would specify a local conservation law, or continuity equation. In the space-time formulation, we expect to find a current j^ν, function of the spatial coordinates, such as $\partial_\nu j^\nu = 0$. Assuming that the spatial components of the quadri-current fall off quickly at spatial infinity and exploiting divergence's theorem,

$$0 = \int d\mathbf{x}\ \partial_\nu j^\nu(x) = \int d\mathbf{x}\ \partial_t \rho(\mathbf{x}, t) + \int d\mathbf{x}\ \nabla \cdot \mathbf{j}(\mathbf{x}, t) = \partial_t Q, \tag{1}$$

where the time coordinate ρ of the current here represents the spatial density of the previous conserved quantity. This statement can be proved, and it's known as the Noether's theorem. Switching to Lagrangian formalism, generally a theory is described by a set of fields $\phi_a(x)$, where a can represent different components of a vectorial field, or different fields. The Lagrangian density $\mathcal{L}$ is a functional of the fields and the respective derivative and completely specifies the (Euler-Lagrange) equations of motion. The same theory can be specified through the action S, defined as the space-time integral of the Lagrangian density. $\mathcal{L}$ is definded up to a total derivative $\partial_\nu K^\nu$, that will only add the boundary term K^ν to the action S and does not touch the equations of motion. In general, a transformation of the fields can be written as $\phi_a(x) \longrightarrow \phi_a(x) + \delta\phi_a(x)$. If we are dealing with a continuous transformation specified by a real value α, we can require it to be infinitesimal simply by taking the limit $\alpha \longrightarrow 0$ and approximate the variation of the Lagrangian density to:

$$\begin{aligned}\delta\mathcal{L} &= \frac{\partial\mathcal{L}}{\partial\phi_a}\delta\phi_a + \frac{\partial\mathcal{L}}{\partial(\partial_\nu\phi_a)}\delta(\partial_\nu\phi_a) \\ &= \partial_\nu\left(\frac{\partial\mathcal{L}}{\partial(\partial_\nu\phi_a)}\delta\phi_a\right) + \left[\frac{\partial\mathcal{L}}{\partial\phi_a} - \partial_\nu\frac{\partial\mathcal{L}}{\partial(\partial_\nu\phi_a)}\right]\delta\phi_a.\end{aligned} \tag{2}$$

The term in square brackets is known to be zero from the Euler-Lagrange equations, therefore we are left with only the first term. We say that a transformation of the fields, $\phi_a(x) \longrightarrow \phi_a(x) + \delta\phi_a(x)$ is a symmetry of the action if the Lagrangian is varied at most by a total derivative $\delta\mathcal{L} = \partial_\nu K^\nu$. If both requirements are satisfied, i.e. ϕ_a satisfy the Euler-Lagrange equations and the transformation

$\delta\phi_a$ is a symmetry of the action, the following continuity equation holds:

$$\partial_\nu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\nu \phi_a)} \delta\phi_a - K^\nu \right) \equiv \partial_\nu j^\nu = 0. \tag{3}$$

This is the local form of the Noether's theorem, which depends only on the symmetry of the action, and is valid for any state that satisfies the equations of motion, hence it does not refer to any particular state. A current will be conserved even if the physical state of a system happens not to share the symmetry of the action, that is, if the symmetry of the action is spontaneously broken.

mezza pagina su esempio con Schrodinger field

2 Gauge freedoms

There are some transformations that leave the action invariant, without any other physical consequence. These transformations are called gauge freedoms. A straightforward example of a gauge freedom is the fact that in any model, the origin of the coordinate system and the orientation and ticks of axes may be freely chosen. Another familiar example of this type of gauge freedom can be found in Maxwell's equations of electromagnetism. Here, the physical observables are the electric and magnetic fields $\mathbf{E}$ and $\mathbf{B}$. These fields do not take arbitrary forms, but are constrained by the Faraday–Maxwell equation and the requirement that the magnetic field is solenoidal:

$$\nabla \times \mathbf{E} + \partial_t \mathbf{B} = 0, \qquad \nabla \cdot \mathbf{B} = 0. \tag{4}$$

These equations fix three of the six components of the electric and magnetic fields. The same constraints can be explicitly enforced by writing $\mathbf{E}$ and $\mathbf{B}$ in terms of the scalar and vector potentials V and $\mathbf{A}$:

$$\mathbf{E} = -\nabla V - \partial_t \mathbf{A}, \qquad \mathbf{B} = \nabla \times \mathbf{A}. \tag{5}$$

Using the scalar and vector potentials, Maxwell's equations take a more convenient form. Introducing the four-potential $A_\mu = (V, \mathbf{A})$, and introducing the kinetic term $-1/4(\partial_\nu A_\mu - \partial_\mu A_\nu)(\partial^\nu A^\mu - \partial^\mu A^\nu)$ in the Lagrangian, Maxwell's equations are automatically satisfied. It can be easily checked that the Lagrangian is left invariant when the relativistic gradient of an arbitrary scalar field $\alpha(x)$ is added to the four-potential:

$$A_\mu \longrightarrow A_\mu + \partial_\mu \alpha(x). \tag{6}$$

This transformation does not affect the physical fields $\mathbf{E}$ and $\mathbf{B}$, it emerges as a symmetry from expressing the Maxwell equations in terms of the four-potential: in fact A_μ has four components, while only three components are needed to completely determine $\mathbf{E}$ and $\mathbf{B}$. At the end of any calculation, however, the final results should not depend on the superfluous degree of freedom which was introduced purely for mathematical convenience. The gauge freedom in Maxwell

electromagnetism actually appears in conjunction with a physical symmetry: the global $U(1)$ symmetry. If j^ν is the corresponding conserved Noehter current, the coupling between the matter field and the gauge field is represented by the addition of the minimal coupling term $A_\nu j^\nu$ to the Lagrangian. This term results invariant under the gauge transformation of Eq. (6) if the additional term $j^\nu \partial_\nu \alpha = \partial_\nu(\alpha j^\nu) - \alpha \partial_\nu j^\nu$ is a total derivative, then if j^ν satisfies the continuity equation.

3 Spontaneous Symmetry Breaking

Spontaneous Symmetry breaking (SSB) occurs when a stable state of the system is not symmetric under a symmetry of the energy functional (Hamiltonian or Lagrangian). This phenomenon always occurs when performing a limit of a parameter that characterize the stability of the system. As an example, we can consider the balancing of a cylinder on a surface. If we sharpen the cylinder at the extremity that touches the surface, decreasing the interface area, it will become more and more difficult to balance the cylinder over the fluctuation force, represented by gravity. As long as the cylinder is perfeclty vertical, both the system and the Hamiltonian share the same rotational symmetry around the axis normal to the surface. On the other hand, balancing a sharp pencil over a table is practically impossible in everyday's life: even with the smallest fluctuation the pencil will fall off, following an arbitrary direction and therefore breaking the rotational symmetry. Every Hamiltonian that can be separated as the sum of the kinetic energies of the components and a potential part suffers from a similar problem. Every macroscopic object, being composed of ions and electrons, should display a similar Hamiltonian. The crucial point to notice about this Hamiltonian, is that it commutes with the operator for total (or centre-of-mass) momentum $\mathbf{P_{tot}}$. This implies that all eigenstates of the crystal are total-momentum eigenstates. Because of Heisenberg's uncertainty principle, states in which the value of the total momentum can be known with certainty must be states in which the centre-of-mass position is entirely unpredictable, resulting in an unlocalized macroscopic object. Separating the interaction part from the collective cente-of-mass motion, this last part is $\mathbf{P}_{tot}/(2N\langle m\rangle)$, where $\langle m\rangle$ is the mean mass of particles and N the number of particles. For every finite number N, $|P=0\rangle$ is the (unique and stable) ground state; in which the center of mass operator $\mathbf{X}_{CoM}$ has infinite uncertainty. The instability of such system comes out in the thermodynamical limit $N \longrightarrow \infty$, thanks to which the energy eigenstates become all degenerate and the probability of finding a superposition of $\mathbf{P}$ states, and then a localized state, is higher and higher. As before, we can simply consider this behaviour by adding to the collective Hamiltonian a small perturbation that tends to localize the object at a certain point, such as the harmonic one $\mu \mathbf{X}^2_{CoM}$. As before, if we manage to cut off every perturbation by doing $\mu \longrightarrow 0$ before the thermodynamical limit, we can still preserve the unlocalized, translationally invariant, ground state; but a more likely situation is that at large, but finite, number of particles N, every

small perturbation will bring the system in a localized state that spontaneously break the translation symmetry of the Hamiltonian. Another famous example is the Heisember ferromagnet/antiferromagnet, a lattice of interacting spins with Hamiltonian

$$H = J \sum_{ij} \mathbf{S}_i \cdot \mathbf{S}_j, \tag{7}$$

where the summation is done over all nearest neighbour and J identifies if the lattic is ferromagetic or antiferromagnetic. Following the previous example, we can define a collective Hamiltonian, function of the total spin $\mathbf{S}_{tot}$ and notice that its eingenvalues become degenerate in the thermodynamical limit: the system can acquire spontaneously $\mathbf{S}_{tot} \neq 0$ with negligible excitation energy. In this case, the Hamiltonian is still symmetric under spin rotation ($SU(2)$ for spin-1/2 particles), but the specific state possess only the $U(1)$ rotation symmetry around the axis specified by $\mathbf{S}_{tot}$.

It's often useful to distinguish the states that break the symmetry. If $U = e^{iQ}$ is a symmetry of the state $|\psi\rangle$, then the expectation value of

$$\langle\psi|[Q, \Psi]|\psi\rangle = 0 \tag{8}$$

for every operator Ψ. If, instead, an operator Ψ that force the commutator to have non-zero expectation value exists, the commutator is called order parameter operator and the expectation value order parameter. Every state that have broken the symmetry of the Hamiltonian U should give a non-zero ordeer parameter and should be an eingenstate of the order parameter operator.

In every situation described before, taking a certain limit the ground state become not only equally favourable with the other excited states, but also strongly unstable. We also notice that the instability is due to the fact that the collective part of the Hamiltonian become negligible with respect to small fluctuations. This behaviour is common to the other thermodynamical potentials, like the free energy F. The collective part of the partition function is

$$Z_{coll} \simeq \int d\mathbf{A}_{tot} \exp\left(\frac{\beta \mathbf{A}_{tot}^2}{N}\right) \propto \sqrt{N}, \tag{9}$$

where $\mathbf{A}_{tot}$ is an operator that commutes with H. The collective part of the free energy F_{coll} should goes with $\ln N$, then $F_{coll}/F \propto \ln N/N \longrightarrow 0$ in the thermodynamical limit, if we want the thermodynamical potential to remain an extrinsic quantity. For systems in which the momentum $\mathbf{k}$ conjugate of position $\mathbf{x}$ is well defined, the kinetic part of the hamiltonian breaks into two pieces:

$$\begin{aligned} H_{kin} &\propto \int d\mathbf{x}\ \mathbf{A}(\mathbf{x})^2 \propto \sum_{kk'} \int d\mathbf{x}\ e^{i(\mathbf{k}-\mathbf{k}')\cdot\mathbf{x}} \mathbf{A}_{\mathbf{k}} \mathbf{A}_{\mathbf{k}'} \\ &= V {\mathbf{A}_{\mathbf{0}}}^2 + V \sum_{\mathbf{k}\neq\mathbf{0}} \mathbf{A}_{\mathbf{k}} \mathbf{A}_{-\mathbf{k}}. \end{aligned}$$

The first part is the collective one: $\mathbf{A_0} = \mathbf{A}_{tot}/V$, while the second part, combined with the potential, describe internal excitations. Also in this case we expect the collective part to become negligible in the thermodynamical limit $V \longrightarrow \infty$.

4 Nambu-Goldstone modes

Noether's theorem applies to every global continuous symmetry of the Hamiltonian, regardless of the fact that the same symmetry is respected by single states. Therefore, every symmetry of the Hamiltonian is associated with a conserved quantity and, in the continuous case, a locally conserved current. As we will see, the local continuity equation impacts on the excitation spectrum of the SSB system at non-zero wave number, and guarantees the appearance of new modes. These modes appear every time a continuous global symmetry is spontaneously broken and when the wave number $\mathbf{k}$ can be defined. The Goldstone theorem states that in absence of long-range interaction a massless (or gapless) mode will appear. If the symmetry is explicitly broken by an external field, the Nambu-Goldstone mode (NG) will be massive. And if the symmetry appears in conjunction with a gauge freedom encoding a long-ranged interaction, the NG mode may couple to the gauge field and develop a gap. Massles NG modes must have a linear dispersion relation $E_k \propto k$ in relativistic systems, but it can be also quadratic in various non-relativistic systems. Type A and type-B NG modes, gap

5 Ginzburg-Landau theory

The SSB mechanism underline a transition between a disordered, symmetric state to an ordered symmetry-breaking state. This specific transition is studied inside the more general framework of phase transitions. Every ordered, symmetry-broken phase has a non-zero expectation value of the order parameter operator. In contrast, the order parameter will be zero in the corresponding symmetric, or disordered, phase. We can therefore distinguish between order and disorder by considering the value of the order parameter. Phase transitions can be happen after a discontinuous variation of the order parameter (first-order phase transitions) or not (second-order phase transitions). The variation is performed with respect to some parameters, such as temperature (or pressure), which specify the phase. In general, equations of state may be expressed in terms of thermodynamic quantities such as pressure, entropy, and so on. These thermodynamic quantities can always be written as derivatives of the free energy. The central idea of Landau theory is that the free energy of any system undergoing a phase transition can be written as a functional of the order parameter, which itself depends on temperature. In a continuous phase transition, the order parameter ψ is guaranteed to be small close to the critical temperature, and one can carry out a Taylor expansion for small ψ in that region. Which

types of terms appear in such an expansion depend strongly on the nature of the order parameter and the symmetries of the system. The equilibrium condition is obtained minimizing the free energy with respect to the order parameter. During the transition, the minimum of the free energy, that identifies the value of the order parameter, shifts towards non-zero values. Ginzburg-Landau (GL) theory treats the phenomenon of superfluidity and superconductivity through a suited free energy functional, although the form of the functional is general and valid for other continuous phase transitions. The theory starts with a complex order parameter ψ. In the uniform case, the free energy can be expanded in series of ψ and ψ^*, keeping in mind the fact that the phase of the order parameter is not observable and hence cannot appear in the free energy. Another common assumption is to keep only the even powers of $\psi\psi^*$, preserving in this way the differentiability of F for $\psi = 0$:

$$F = \alpha|\psi|^2 + \frac{\beta}{2}|\psi|^4 + \mathcal{O}(|\psi|^6), \tag{10}$$

where F is referenced to the unordered phase. $\beta > 0$ is necessary to keep the free energy bounded, avoiding it to take arbitrarily negative values. For $\alpha \geq 0$ the minimum is found at $\psi = 0$. Defining the temperature parameter below which the system experiences the phase transition as T_c, in this regime we are at $T > T_c$. When $\alpha < 0$, the minimum is found at (Fig. (1)):

$$\frac{\partial F}{\partial|\psi|} = 2|\psi|(\alpha + \beta|\psi|^2) = 0, \tag{11}$$

where $|\psi| = \sqrt{-\alpha/\beta}$ is the minimum condition. Since α changes sign during the transition, we can assume (at first order) that $\alpha \sim (T - T_c)$, so that

$$\psi \sim (T - T_c)^{1/2} e^{i\phi}. \tag{12}$$

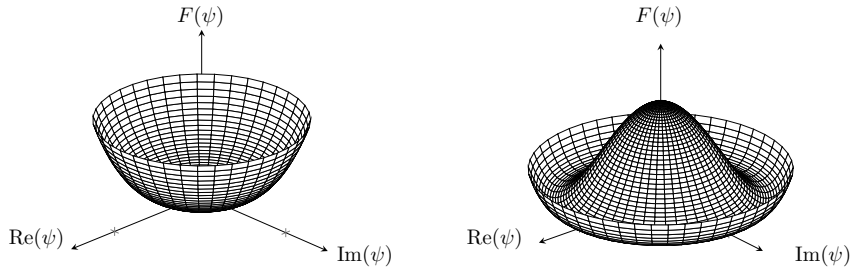
As it turns out, the scale invariance of the system at a continuous phase transition guarantees the behavior of the order parameter close to the critical temperature to always be of the form $(T - T_c)^a$.

6 GL for neutral superfluids

More generally, assuming ψ non-uniform, we can find the equilibrium equation of state exaclty like before, adding a gradient term. The free energy of Eq. (11) become a free energy density, while F now is the integral of the density:

$$F[\psi] = \int d^3r \left[\alpha|\psi|^2 + \frac{\beta}{2}|\psi|^4 + \gamma|\nabla\psi|^2\right]. \tag{13}$$

The gradient term is added at the first allowed order, considering small variations; in addition, we must take $\gamma > 0$ to avoid arbitrarily negative free energy



(a) $T > T_c$ (b) $T < T_c$

Figure 1: Plot of free energy profile of Fig. (10). The phase transition occurs from unordered state (left) to the ordered one (right). The order parameter acquire an arbitrary phase, breaking the $U(1)$ symmetry. After the phase transition, amplitude fluctuation around the minima require some energy (massive), while phase fluctuation connect only degenerate states (massless NG mode).

density increasing the non-uniformity. If F is minimum for ψ_0, then $\delta F = 0$ keeping the variation $\delta\psi$ of the order parameter small:

$$\delta F = \int d^3r \left[\alpha\psi_0 + \beta\psi_0|\psi_0|^2\right] \delta\psi^* + \gamma\nabla\psi_0 \cdot \nabla\delta\psi^* + \mathcal{O}(\delta\psi^{*2}, \nabla\delta\psi^{*2}). \quad (14)$$

Only $\delta\psi^*$ and $\nabla\delta\psi^*$ are explicitly expanded, because from $\delta\psi$ we derive the same (conjugate) equation. Integratin by parts the term containing the gradient, we would have $\nabla\psi_0 \cdot \delta\psi^* \longrightarrow 0$ because we cannot allow spatial variations at the domain boundaries ($\nabla\psi_0 \longrightarrow 0$ at $||\mathbf{r}|| \longrightarrow \infty$). The remaining term is

$$\delta F[\delta\psi^*] \simeq \int d^3r \left[\alpha\psi_0 + \beta\psi_0|\psi_0|^2 - \gamma\nabla^2\psi_0\right] \delta\psi^*. \quad (15)$$

The variation of the free energy is guaranteed to be zero (no matter of $\delta\psi^*$) if the term in square brackets is zero, obtaining in this way a non-linear generalization of Schrödinger equation, in which $\gamma = \hbar^2/2m$:

$$-\frac{\hbar^2}{2m}\nabla^2\psi_0 + \alpha\psi_0 + \beta|\psi_0|^2\psi_0 = 0. \quad (16)$$

Up to now we have neglected fluctuations of the order parameter (always existing at $T > 0$), that is we have found a mean-field solution. While the equilibrium imposes F to not depend on fluctuations $\delta\psi$ of the first order, now we want to consider fluctuations $\delta\psi$ up to the second order. From here it's more convenient to work on the $k-$space, where the fluctuation is decomposed in each normal mode $\delta\psi_{\mathbf{k}}$:

$$\delta\psi(r) = \frac{1}{\sqrt{V}} \sum_k \delta\psi_k e^{ik\cdot r}. \quad (17)$$

The structure of the functional depends on the mean-field solution, which depends on the phase. We are then dealing with two cases:

- $\boldsymbol{T > T_c}$: the mean-field solution is $\psi_0 = 0$, it is sufficient to substitute $\psi = \delta\psi$ in Eq. (13) and to neglect the fourth order term containing β. The energy functional is then written as:

$$F[\delta\psi] \simeq \int d^3r \left[\alpha|\delta\psi(r)|^2 + \frac{\hbar^2}{2m}|\nabla\delta\psi(r)|^2\right], \tag{18}$$

and on the basis $\delta\psi_k$:

$$F[\delta\psi] \simeq \frac{1}{V}\sum_{k,k'}\left[\alpha + \frac{\hbar k k'}{2m^*}\right]\delta\psi_k^*\delta\psi_{k'}\int d^3r\, e^{i(k'-k)r}, \tag{19}$$

where the last integral is $V\delta_{kk'}$. Finally:

$$F[\delta\psi] \simeq \sum_k\left[\alpha + \frac{\hbar^2k^2}{2m^*}\right]\delta\psi_k^*\delta\psi_k \tag{20}$$

- $\boldsymbol{T < T_c}$: the only constrain in the mean-field solution is that of Eq. (16). Writing $\psi = (\psi_0 + \delta\psi)e^{i(\phi+\delta\phi)}$ we seek the dispersion relation of the two (real) modes $\delta\psi$ and $\delta\phi$. The energy functional is:

$$\begin{aligned} F[\psi] &\simeq \int d^3r \left[\alpha\delta\psi^2 + 3\beta\psi_0^2\delta\psi^2 + \frac{\hbar^2}{2m^*}|\nabla((\psi_0+\delta\psi)e^{i(\phi+\delta\phi)}|^2\right] \\ &\simeq \int d^3r \left[(\alpha + 3\beta\left(-\frac{\alpha}{\beta}\right))\delta\psi^2 + \frac{\hbar^2}{2m^*}|e^{i\phi}(\nabla\delta\psi + i\psi_0\nabla\delta\phi)|^2\right] \\ &\simeq \int d^3r \left[-2\alpha\delta\psi^2 + \frac{\hbar^2}{2m^*}((\nabla\delta\psi)^2 + (\nabla\delta\phi)^2)\right], \end{aligned} \tag{21}$$

and with the same procedure, we obtain the (Fourier transformed) normal modes:

$$F[\delta\psi, \delta\phi] = \sum_k\left[\left(-2\alpha + \frac{\hbar^2k^2}{2m^*}\right)\delta\psi_k^2 - \frac{\alpha\hbar^2k^2}{2m^*\beta}\delta\phi_k^2\right] \tag{22}$$

We have demonstrated what it's clear from Fig. (1): before transition, we expect two coupled normal modes that are gapped: the free energy increases while moving around the unique $\psi_0 = 0$ equilibrium position, hence a minimum activation energy is required. On the contrary, after the phase transition, amplitude fluctuations decouple from phase fluctuations: the first continue to be massive, the last have a non-massive dispersion relations.

7 GL for superconductors

In the Ginzburg-Landau theory, superconductors are nothing more than superfluid with charge, the main differences are in the energy functional: in presence of an electromagnetic field, we should add the kinetic term of the field (energy density of the field $B^2 + E^2$) and the interaction term is fully considered substituting the momentum $-i\hbar\nabla$ with $-i\hbar\nabla - (q/c)\boldsymbol{A}$. Neglecting the electrostatic field ($E = 0$ and the electrostatic potential $\phi(r) = 0$) and using gaussian units,

$$F = \int d^3r \left[\alpha|\psi|^2 + \frac{\beta}{2}|\psi|^4 + \frac{1}{2m}\left|\left(\frac{\hbar}{i}\nabla - \frac{q}{c}\boldsymbol{A}\right)\psi\right|^2 + \frac{(\nabla\times\boldsymbol{A})^2}{8\pi}\right]. \tag{23}$$

This time we expect two equilibrium equations, the first (Ginzburg-Landau) equation is obtained varying ψ and it's formally identical to Eq.(16), with the modified momentum:

$$\frac{1}{2m^*}\left(\frac{\hbar}{i}\nabla - \frac{q}{c}\boldsymbol{A}\right)^2\psi + \alpha\psi + \beta|\psi|^2\psi = 0 \tag{24}$$

The second equation of state is obtained minimizing with respect to the electromagnetic field $\mathbf{A}$, keeping as before only the linear terms $\delta\mathbf{A}$. The gradient term yields the following contribution:

$$-\frac{q}{c}\psi\delta\boldsymbol{A}\cdot\left[\left(\frac{\hbar}{i}\nabla\psi\right)^* - (\frac{q}{c}\boldsymbol{A}\psi)^*\right] - \psi^*\delta\boldsymbol{A}\cdot\left[\frac{\hbar}{i}\nabla - \frac{q}{c}\boldsymbol{A}\right] \tag{25}$$

which becomes

$$-\frac{i\hbar q}{c}\delta\boldsymbol{A}\cdot(\psi\nabla\psi^* - \psi^*\nabla\psi) + 2\frac{q^2}{c^2}|\psi|^2\delta\boldsymbol{A}\cdot\boldsymbol{A}. \tag{26}$$

The linear term extracted from B^2 is

$$2\boldsymbol{B}\cdot(\nabla\times\delta\boldsymbol{A}) = 2(\nabla\times\boldsymbol{B})\cdot\delta\boldsymbol{A}, \tag{27}$$

defining $\boldsymbol{B} = \nabla\times\boldsymbol{A}$ and from $\nabla\cdot\boldsymbol{B} = 0$.

$$\delta F[\delta\boldsymbol{A}] = \int d^3r\left[-\frac{qi\hbar}{2m^*c}(\psi\nabla\psi^* - \psi^*\nabla\psi) + \frac{q^2}{m^*c^2}|\psi|^2\boldsymbol{A} + \frac{(\nabla\times\boldsymbol{B})}{4\pi}\right]\cdot\delta\boldsymbol{A}. \tag{28}$$

The Ampère's Law states that the last term is a current $\boldsymbol{j}/c$; the effective masses and charges are doubled, because from the microscopic theory it is found that the particles inside the superfluid are bound states of two electrons. The second GL equation is then:

$$\boldsymbol{j} = \frac{i\hbar e}{2m}\left[\psi^*\nabla\psi - \psi\nabla\psi^*\right] - \frac{2e^2}{mc}|\psi|^2\boldsymbol{A}, \tag{29}$$

and finally, supposing spatial phase variation more sensitive than the amplitude one, the previous equation reduces to:

$$\boldsymbol{j} = -\frac{\hbar e n_s}{2m}\nabla\phi - \frac{n_s e^2}{mc}\boldsymbol{A}. \tag{30}$$

This conserved current can be mantained by electromagnetic fields, but also by spatial varations of the phase of the condensate: a supercurrent can flow even in the absence of external fields. It's important to notice that this is an equilibrium equation, therefore the current flows even once minimum energy has been reached.

8 Anderson-Higgs Mechanism

Now we analyze fluctuations from the equilibrium position, in this case also electromagnetic field fluctuations are present. As before, we have two different cases, because of the different mean-field solution.

- $\boldsymbol{T > T_c}$: as before, we substitute $\delta\psi$ to ψ. Naturally, the minimum of the functional is reached when wither $\mathbf{A} = 0$. The kinetic term simplifies to

$$\left|\left(\frac{\hbar}{i}\nabla - \frac{q}{c}\delta\boldsymbol{A}\right)\delta\psi\right|^2 \simeq \hbar^2|\nabla\delta\psi|^2. \tag{31}$$

  The Fourier transform should be equal to that of the Eq. (20), with the addition of the energy density of the field

$$F[\delta\psi_k, \delta\boldsymbol{A_k}] \simeq \sum_k \left[\left(\alpha + \frac{\hbar^2 k^2}{2m^*}\right)\delta\psi_k^*\delta\psi_k + \frac{|\boldsymbol{k}\times\delta\boldsymbol{A_k}|^2}{8\pi}\right]. \tag{32}$$

  The density term B^2 is always present in functionals where electromagnetic fluctuations are considered. Due to the nature of the vector product, the oscillations are always transversal with reference to the direction of propagation $\boldsymbol{k}$. The transform is then rewritten highlighting this fact, in the following way:

$$F[\delta\psi_k, \delta\boldsymbol{A_k}] \simeq \sum_k \left[\left(\alpha + \frac{\hbar^2 k^2}{2m^*}\right)\delta\psi_k^*\delta\psi_k + \frac{k^2}{8\pi}\delta\boldsymbol{A}_{\boldsymbol{k}}^{\perp}\delta\boldsymbol{A}_{\boldsymbol{k}}^{\perp *}\right]. \tag{33}$$

  Exactly like in the superfluid case, the gap of order parameter normal modes is reduced decreasing the temperature, vanishing at the critical temperature T_c. The normal modes of the external electromagnetic field are naturally gapless: the field penetrates in the material without losing intensity. In the normal state, up to second order, there is no coupling between electromagnetic and order parameter's fluctuation, as represented by the fact that there are no mixed modes.

- $\boldsymbol{T < T_c}$: As before we are dealing separately with amplitude and phase fluctuations. The kinetic term is:

$$\begin{aligned}&\left|-i\hbar\nabla\psi_0 - i\hbar\nabla\delta\psi + (\psi_0+\delta\psi)\left(\hbar\nabla\phi - \frac{q}{c}\delta\boldsymbol{A}\right)\right|^2 \\ &\simeq \hbar^2(\nabla\delta\psi)^2 + \hbar^2\psi_0^2\left|\nabla\phi - \frac{q}{\hbar c}\delta\boldsymbol{A}\right|^2 \end{aligned} \tag{34}$$

while the electromagnetic density term is keep untouched, displaying only transversal fluctuations. This time mixed modes will appear, and the kinetic term in $\delta\boldsymbol{A}\delta\boldsymbol{A}^*$ will be responsible for the appearance of a gap in the dispersion relation of electromagnetic fluctuations (massive photon). The appearance of a phase has certainly broken the global $U(1)$ symmetry, but the resulting energy functional should be still be symmetric under the typical gauge transformation of the electromagnetic field:

$$\begin{aligned}\delta\boldsymbol{A} &\longrightarrow \delta\boldsymbol{A} + \frac{\hbar c}{q}\nabla\phi(r) \\ \psi &\longrightarrow e^{i\phi(r)}\end{aligned} \tag{35}$$

In this way, performing the right gauge transformation, the phase in Eq. (34) disappears: phase fluctuations are incorporated in the electromagnetic ones. The transform of the functional is finally:

$$F[\delta\psi,\delta\boldsymbol{A}] \simeq \sum_k \left[\left(-2\alpha + \frac{\hbar^2k^2}{2m^*}\right)\delta\psi_k^2 - \frac{\alpha q^2}{2\beta c^2 m^*}\delta\boldsymbol{A}_{\boldsymbol{k}}^*\delta\boldsymbol{A}_{\boldsymbol{k}} + \frac{k^2}{8\pi}\delta\boldsymbol{A}_{\boldsymbol{k}}^{\perp *}\delta\boldsymbol{A}_{\boldsymbol{k}}^{\perp}\right].$$

Dividing transversal and longitudinal electromagnetic modes, the previous expression can be rewritten as

$$-\frac{\alpha q^2}{2\beta c^2 m^*}\delta\boldsymbol{A}_{\boldsymbol{k}}^{\parallel *}\delta\boldsymbol{A}_{\boldsymbol{k}}^{\parallel} \tag{36}$$

for longitudinal modes and

$$\left(-\frac{\alpha q^2}{2\beta c^2 m^*} + \frac{k^2}{8\pi}\right)\delta\boldsymbol{A}_{\boldsymbol{k}}^{\perp *}\delta\boldsymbol{A}_{\boldsymbol{k}}^{\perp} \tag{37}$$

for the transversal ones.

Unlike the normal state, the symmetry breaking state is characterized by the existence of longitudinal modes. Also, both modes acquire a gap, which coincide in the language of the particle physics to a massive field vector, in which the mass is given by

$$-\frac{\alpha}{\beta}\frac{8\pi q^2}{2m^*c^2} = |\psi_0|^2\frac{8\pi(4e^2)}{2(2m)c^2} = \frac{4\pi e^2 n_s}{mc^2} = \frac{1}{\lambda_L^2}, \tag{38}$$

where n_s is the fraction of superfluid density and λ_L is the characteristic penetration depth after which the external field can be neglected because of the (previously known empirically as Meissner effect) exponential damping. This mechanism is also known as Anderson-Higgs mechanism.